

esteban romero (he / him)

Exploring how systems of play and interaction connect us to new worlds.

Publications, Conferences, and Awards

Conference Speaker: YouQuantified June, 2024
@ 5th International Mobile Brain / Imaging Conference in Piran, Slovenia ([Link](#))

TEDx Conference Speaker: Pictures From the Mind June, 2024
@ Tecnológico de Monterrey, Guadalajara ([Webpage Link](#))

Publication: Wearable Biosensor Technologies in Education April, 2024
@ MDPI Sensors Journal ([Link](#))

Conference Speaker: Techno Futuristic Dreams of Virtual Utopias April, 2026
@ GS1 New York University ([Link](#))

Award: A Thousand Word Speller April, 2023
@ Virtual BR4IN.io Hackathon in the Programming & Arts Category ([Link](#))

Projects

Developer & Researcher: MindHive & YouQuantified July, 2023 – Present
@New York University ([Project Website](#))

- Led the development and design of the YouQuantified app, a web-based application aimed at teaching students how to visualize and interact with the data produced by their bodies (biosensors & face tracking) in real-time with models that run in the client.
- Co-led implementations of the tool in high school classrooms across New York City and Boston.
- Helped in the design and implementation of data analysis and project planning tools for MindHive, a platform being used by over 500 students to plan, build, and conduct cognitive science studies.

Data Visualization & Analysis: XenBox July, 2023 – June, 2024
@New York University MindHive Lab ([RollingStone Article](#))

- Worked in collaboration with Mike Gordon from Phish and Bob Weir from The Grateful Dead to develop a data analysis pipeline for a dyadic jamming session where heart-rate, skin conductance, and EEG data were being collected in real-time.
- Designed a custom analysis pipeline to synchronize all data streams with pre-recorded video, display summary statistics, and transform streams.

Playful Intervention: Stop To Play

Fall, 2025

@ New York University

- Co-developed and playtested a “playful intervention” designed for the NYC Subway. This was an online multiplayer incremental game for “social non-social play” or simple anonymous interactions between strangers in their commutes.

Creative Lead & Game Designer: Mynd Music

September, 2022 – January, 2023

@ Tecnológico de Monterrey & Laboratorio de Arte AC

- Designed and developed a Unity-based VR experience with real-time auditory feedback corresponding obtained from correlated users’ affective states. This project was presented and demoed at the incMTY 2022 festival. (Demo & Paper)

Local Production: Mutual Wave Machine

September, 2022 – January, 2023

@ Tecnológico de Monterrey, Guadalajara

- Led the 3D modeling, Processing coding, and implementation of a version of the Mutual Wave Machine, by professor Suzanne Dikker. This was a 2-person art installation where people could sit in front of each other and through projectors behind them, visualize their “shared synchrony” as measured by EEG headsets.
- This project was showcased at Tecnológico de Monterrey during student fairs through the *wavesense* student club.

Developer: Beat Motion

Spring, 2026

@ Guadalajara, México

- This was a Kinect-based MIDI controller used to translate limb positions into knobs and triggers for music production and performance.

Collaborator: New York City Hall of Science

Spring, 2025

@ New York University, NYSCI

- Ran a series of educational workshops and helped in the co-design of tabletop activities and games with their “Explainers”. We explored technologies for biosensing and real-time machine learning models for facial and emotion recognition from video streams.
- Collaborated with Columbia’s ARNI lab for the brainstorming and design of these activities.

Digital Game Design: Get It Together

Spring, 2026

@ New York University [[Itch Page](#)]

- Co-designed a local multiplayer party-combat game developed by a team of four game designers.
- Led the design and implementation of the VFX and custom shaders. Helped in coding the gameplay loop and custom physics.

- Shown at NYU's Game Center 2026 Spring Showcase.

Analogue Game Design: Sweet & Sour

Fall, 2025 – Spring, 2026

@ New York University

- Co-designed a 4 to 6 player role-playing game about family dynamics.

Analogue Game Design: Midnight Gala

Spring, 2026

@ New York University

- Polished, playtested, and co-designed a large-scale (12+ people) social deduction party game for events & gatherings.

Member of Showcase Committee

Spring, 2026

@ New York University

- Helped in the organization of our annual Game Center Spring Showcase with more than 800 attendees. Specifically, I was part of the team involved in branding & printed materials.

Assistant: Lincoln Center's Game Making Create-athon

April, 2026

@ New York University [[Event Link](#)]

- Assisted in running a community-outreach event that allowed children, families, and adults to get hands-on experience with analogue game design by modding and creating their own games.

Assistant: THE TEST

October, 2025

@ New York University

- Assisted in running the initial playtest for a large-scale social game and artistic experience designed by Eric Zimmerman.

Vice-President and Co-Founder: wavesense Student Club

January 2022 – June 2023

@ Tecnológico de Monterrey, Guadalajara

- Led a 30+ member student club dedicated to neuroscience workshops, science outreach, and research.
- Worked on biofeedback-based games and installations for events and hackathons for the student community.

Research Affiliate: NT Lab

March 2022 – June 2023

@ Tecnológico de Monterrey, Guadalajara

- Researched on motor rehabilitation treatments using a neuroprosthesis.

Research Affiliate: BRAIN Tec

August 2022 – June 2023

@ Tecnológico de Monterrey, Monterrey

- Co-developed software for the real-time data collection, recording, and preprocessing of a brain-to-brain synchrony study in education.
- Co-wrote and published a review on the use of Wearable Devices in Education.

Skills

Game Development & Interactive Systems

Unity (C#) • Biofeedback-driven games with PPG and EEG data • Tech Art & HLSL Shader Coding • Motion Tracking with OpenCV, MediaPipe and the Xbox Kinect SDK

Biomedical Engineering

Real-time physiological data processing • Experimental design & data collection • Motion Capture & Data Processing for Biomechanics

Design & UX

Designing UX/UI for Educational Settings in Figma • Creative data visualization & coding in P5.js • Designing Family Museum Installation

Education

MFA in Game Design

September, 2025 – May, 2027

@ New York University

BS in Biomedical Engineering

August, 2020 – June, 2024

@ Tecnológico de Monterrey | 4.0 GPA & Top 5% of Class